

TCS3351 CRYPTOGRAPHY AND DATA SECURITY
Tutorial 8 - Modern Asymmetric Key Ciphers: Principles

1. State the major differences between symmetric-key and asymmetric-key cryptosystems.
2. With the help of a diagram, explain how the public-key cryptosystem can be used to provide authentication/digital signature.
3. With the help of a diagram, explain how the public-key cryptosystem can be used to provide both confidentiality and authentication.
4. Briefly explain the One-way and Trap-door one-way functions.

References:

1. Forouzan, B. A. & Mukhopadhyay, D. (2015). Cryptography & Network Security (3rd ed.). McGraw-Hill.
2. William Stallings, “Cryptography and Network Security – Principles and Practice”, Seventh Edition, Prentice Hall, 2017